

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
Regular/Supplementary Winter Examination – 2024			
Course: B. Tech. Branch : Electronics and Computer Engineering/Electronics and Computer Science			
Subject Code & Name: Software Engineering (BTECPE503D)		Semester : Fifth	
Max Marks: 60		Date: 11/02/2025	
		Duration: 3 Hr.	
Instructions to the Students:			
1. Each question carries 12 marks.			
2. Question No. 1 will be compulsory and include objective-type questions.			
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.			
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.			
5. Use of non-programmable scientific calculators is allowed.			
6. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Software is defined as-----		
	a. Set of programs, documentation & configuration of data		
	b. set of programs	L-1/CO1	1
	c. documentation and configuration of data		
	d. None of the mentioned		
	What is Software Engineering?		
	a. Designing a software		
	b. Testing a software	L-1/CO1	1
	c. Application of engineering principles to the design a software		
	d. None of the above		
3	_____ is a software development activity that is not a part of software processes.		
	a. Validation		
	b. Specification	L-1/CO1	1
	c. Development		
	d. Dependence		
	Define Agile scrum methodology.		
	a. Project management that emphasizes incremental progress		
	b. Project management that emphasizes Decremental progress	L-1/CO2	1
	c. Project management that emphasizes neutral progress		
	d. Project management that emphasizes no progress		

5	What is a Functional Requirement?	L-1/CO2	1
	a. Specifies the tasks the program must complete		
	b. Specifies the tasks the program should not complete		
	c. Specifies the tasks the program must not work		
	d. All of the mentioned		
6	Which is not a step of Requirement Engineering?	L-1/CO2	1
	a. Requirements elicitation		
	b. Requirements analysis		
	c. Requirements design		
	d. Requirements documentation		
7	The _____ model helps in representing the system's dynamic behavior.	L-1/CO3	1
	a. Object Model		
	b. Context Model		
	c. Behavioral Model		
	d. Data Model		
8	Which design identifies the software as a system with many components interacting with each other?	L-1/CO3	1
	a. Architectural design		
	b. High-level design		
	c. Detailed design		
	d. Both B & C		
9	If the objects focus on the problem domain, then we are concerned with _____.	L-1/CO4	1
	a. Object Oriented Analysis		
	b. Object Oriented Design		
	c. Object Oriented Analysis and Design		
	d. None of the above		
10	Which of the following is not open source software?	L-1/CO4	1
	a. LibreOffice		
	b. Microsoft Office		
	c. GNU image manipulation		
	d. MySQL		

11	Software testing is:-----		
	a. the process of demonstrating that errors are not present b. the process of establishing confidence that a program does what it is supposed to do c. the process of executing a program to show it is working as per specifications d. the process of executing a program with the intent of finding errors	L-1/CO5	1
12	Which of the following testing is the part of non-functional testing?		
	a. Unit Testing b. Performance Testing c. System Testing d. Integration Testing	L-1/CO5	1
Q. 2	Solve the following.		12
A)	Describe incremental approach for software development.	L-2/CO1	6
B)	Explain in brief characteristics of software.	L-3/CO1	6
Q. 3	Solve the following.		12
A)	Define Agility. List any five principles of Agility.	L-2/CO2	6
B)	Explain and compare functional and nonfunctional requirements in Software Engineering.	L-3/CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What do you mean by system modeling? Explain Context Model.	L-3/CO3	6
B)	Discuss the layered architecture pattern with four layers.	L-2/CO3	6
C)	Explain Pipe and filter architecture.	L-3/CO3	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the process of Object-oriented design using UML.	L-3/CO4	6
B)	Describe essential elements of Design patterns.	L-2/CO4	6
C)	Write a short note on open source development.	L-1/CO4	6

Q. 6	Solve Any Two of the following.		12
A)	Define software testing. Which are the different benefits of software testing?	L-2/CO5	6
B)	What is the difference between validation and verification?	L-2/CO5	6
C)	Explains safety requirements and safety cases in software testing.	L-3/CO5	6
*** End ***			